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Carlos' lectures on Hilbert spaces, Dirac formalism, discrete representations, observables, dual spaces.

~ Position & momentum representations

As we already used, the representation of the wave function $|\psi\rangle$ on the position rep. is given by

$$\psi(x) = \langle x | \psi \rangle$$

Analogously, in the momentum rep. we have

$$\varphi(p) = \langle p | \psi \rangle,$$

when we assume that both basis are complete:

$$\int dp |p\rangle \langle p| = 1, \quad \int dx |x\rangle \langle x| = 1$$

Furthermore, they are orthonormal

$$\int dp \langle p | p' \rangle = \delta^0(p - p'), \quad \langle x | x' \rangle = \delta^0(x - x')$$

Clearly, to obtain $\varphi(p)$ in terms of $\psi(x)$

$$\begin{aligned} \varphi(p) &= \langle p | \psi \rangle = \int dx \langle p | x \rangle \langle x | \psi \rangle \\ &= \int dx \langle p | x \rangle \psi(x) \end{aligned}$$

and conversely

$$\begin{aligned} \psi(x) &= \langle x | \psi \rangle = \int dp \langle x | p \rangle \langle p | \psi \rangle \\ &= \int dp \langle x | p \rangle \varphi(p) \end{aligned}$$

where $\langle p | x \rangle = \langle x | p \rangle^*$ is still unknown.

Combining both expressions, we find

$$\begin{aligned}\psi(x) &= \int dp \langle x|p \rangle \int dx' \langle p|x' \rangle \psi(x') \\ &= \int dx' \psi(x') \int dp \langle p|x' \rangle \langle x|p \rangle\end{aligned}$$

$$\Leftrightarrow \int dp \langle x|p \rangle \langle p|x' \rangle = \delta^D(x-x') \quad (\text{use } D=1)$$

There exists a ~~for~~ similar integral in the Fourier transform

$$\delta(x-x') = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2\pi}} \int dp e^{-ip \cdot (x-x')}$$

which can be rewritten in Dirac's normalization $p \rightarrow p/\hbar$ as

$$\delta(x-x') = \left(\frac{1}{\sqrt{2\pi\hbar}} \right)^2 \int dp e^{-ip(x-x')/\hbar}$$

Thus, one can conjecture that

$$\langle p|x \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{\pm i p x / \hbar} \quad \text{minus sign is conventional}$$

For $D=3$ we have

$$\langle \vec{p} | \vec{x} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3p e^{-i\vec{p} \cdot \vec{x} / \hbar}$$

It follows that in $D=3$

$$\psi(\vec{x}) = \langle \vec{x} | \psi \rangle = \int d^3p \langle \vec{x} | \vec{p} \rangle \langle \vec{p} | \psi \rangle = \int d^3p \frac{1}{(2\pi\hbar)^{3/2}} e^{i\vec{p} \cdot \vec{x} / \hbar} \psi(\vec{p})$$

$$\psi(\vec{p}) = \langle \vec{p} | \psi \rangle = \int d^3x \langle \vec{p} | \vec{x} \rangle \langle \vec{x} | \psi \rangle = \int d^3x \frac{1}{(2\pi\hbar)^{3/2}} e^{-i\vec{p} \cdot \vec{x} / \hbar} \psi(\vec{x})$$

Notice that $\langle \psi | \psi \rangle$ (and thus the length of a state) is independent of the representation (47)

$$\begin{aligned}\langle \psi | \psi \rangle &= \int d^3x \langle \psi | \vec{x} \rangle \langle \vec{x} | \psi \rangle = \int d^3x \psi(\vec{x})^* \psi(\vec{x}) \\ &= \int d^3p \langle \psi | \vec{p} \rangle \langle \vec{p} | \psi \rangle = \int d^3p \psi(\vec{p})^* \psi(\vec{p})\end{aligned}$$

The matrix element of an operator $\hat{A}$ in these reps. is related by

$$\begin{aligned}\hat{A}(\vec{p}, \vec{p}') &\equiv \langle \vec{p} | \hat{A} | \vec{p}' \rangle = \int d^3x \int d^3x' \frac{1}{(2\pi\hbar)^3} \langle \vec{p} | \vec{x} \rangle \langle \vec{x} | \hat{A} | \vec{x}' \rangle \langle \vec{x}' | \vec{p}' \rangle \\ &= \frac{1}{(2\pi\hbar)^3} \int d^3x \int d^3x' e^{\frac{i}{\hbar}(\vec{p} \cdot \vec{x} - \vec{p}' \cdot \vec{x}')} \underbrace{\langle \vec{x} | \hat{A} | \vec{x}' \rangle}_{\hat{A}(\vec{x}, \vec{x}')}\end{aligned}$$

Another important observation is that an operator is diagonal in its own representation, i.e. in $D=1$

$$\langle \vec{p}' | \hat{p} | \vec{p}'' \rangle = \vec{p}' \delta(\vec{p}' - \vec{p}'')$$

$$\langle \vec{x}' | \hat{x} | \vec{x}'' \rangle = \vec{x}' \delta(\vec{x}' - \vec{x}'')$$

It follows that (see Cohen-Tannoudji p. 182)

$$\langle \vec{x}' | F(\hat{x}) | \vec{x}'' \rangle = F(\vec{x}') \delta(\vec{x}' - \vec{x}'')$$

$$\langle \vec{p}' | F(\hat{p}) | \vec{p}'' \rangle = F(\vec{p}') \delta(\vec{p}' - \vec{p}'')$$

- Schrödinger's equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

time-dependent

$$E |\psi\rangle = \hat{H} |\psi\rangle$$

stationary

$$\hat{H} \equiv \frac{\hat{p}^2}{2m} + \hat{V}(\vec{x})$$

Stationary case in $\{|\vec{x}\rangle\}$ up

$$\begin{aligned} E \langle \vec{x} | \psi \rangle &= \langle \vec{x} | \frac{\hat{p}^2}{2m} | \psi \rangle + \langle \vec{x} | \hat{V}(\vec{x}) | \psi \rangle \\ &= -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{x}) + \int d^3x' \langle \vec{x} | \hat{V}(\vec{x}) | \vec{x}' \rangle \langle \vec{x}' | \psi \rangle \\ &= -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{x}) + V(\vec{x}) \int d^3x' \delta^3(\vec{x} - \vec{x}') \psi(\vec{x}') \\ &= -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{x}) + V(\vec{x}) \psi(\vec{x}) \end{aligned}$$

In momentum sp.

$$\begin{aligned} E \langle \vec{p} | \psi \rangle &= \langle \vec{p} | \frac{\hat{p}^2}{2m} | \psi \rangle + \langle \vec{p} | \hat{V}(\vec{x}) | \psi \rangle \\ &= \int d^3p' \langle \vec{p} | \frac{\hat{p}^2}{2m} | \vec{p}' \rangle \langle \vec{p}' | \psi \rangle + \int d^3p' \langle \vec{p} | \hat{V}(\vec{x}) | \vec{p}' \rangle \langle \vec{p}' | \psi \rangle \end{aligned}$$

$$\langle \vec{p} | \frac{\hat{p}^2}{2m} | \vec{p}' \rangle = \frac{\vec{p}^2}{2m} \delta^3(\vec{p} - \vec{p}')$$

$$\begin{aligned} \langle \vec{p} | \hat{V}(\vec{x}) | \vec{p}' \rangle &= \int d^3x \int d^3x' \langle \vec{p} | \vec{x} \rangle \langle \vec{x} | \hat{V}(\vec{x}) | \vec{x}' \rangle \langle \vec{x}' | \vec{p}' \rangle \\ &= \int d^3x \int d^3x' \frac{1}{(2\pi\hbar)^3} e^{-i\vec{p}\cdot\vec{x}/\hbar} V(\vec{x}) \delta^3(\vec{x} - \vec{x}') e^{i\vec{p}'\cdot\vec{x}'/\hbar} \\ &= \int d^3x \frac{1}{(2\pi\hbar)^3} e^{-i\vec{p}\cdot\vec{x}/\hbar} V(\vec{x}) \int d^3x' \delta^3(\vec{x} - \vec{x}') e^{i\vec{p}'\cdot\vec{x}'/\hbar} \end{aligned}$$

shifting theorem
Fourier transf.

$$\begin{aligned} &= \int d^3x \frac{1}{(2\pi\hbar)^3} e^{-i\vec{x}\cdot(\vec{p}-\vec{p}')/\hbar} V(\vec{x}) \\ &= \frac{1}{(2\pi\hbar)^{3/2}} V(\vec{p} - \vec{p}') \quad \text{with} \quad V(\vec{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3x V(\vec{x}) e^{-i\vec{p}\cdot\vec{x}/\hbar} \end{aligned}$$

$$\Rightarrow E \psi(\vec{p}) = \frac{\vec{p}^2}{2m} \psi(\vec{p}) + \int d^3p' \frac{1}{(2\pi\hbar)^{3/2}} V(\vec{p} - \vec{p}') \psi(\vec{p}')$$

Observations:

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1) Clearly, for a free particle in $D=1$

$$V(x)=0 \Rightarrow V(p)=0$$

In position-space, the solutions are

$$\psi(x) = A e^{ikx} + B e^{-ikx} \quad k^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow \boxed{\psi(p) = \int dx \psi(x) e^{-ipx/\hbar}} = \frac{A}{\sqrt{2\pi\hbar}} \int dx e^{-ix(\frac{p}{\hbar} - k)} + \frac{B}{\sqrt{2\pi\hbar}} \int dx e^{-ix(\frac{p}{\hbar} + k)}$$
$$= A' \delta(p - k\hbar) + B' \delta(p + k\hbar)$$

In mom.-sp. the Schrödinger eq. reads

$$\left(\frac{p^2}{2m} - E\right) \psi(p) = 0 \rightarrow \psi(p) = 0 \text{ everywhere except when}$$
$$p^2 = 2mE$$

$$\Rightarrow \psi(p) = A \delta(p - \sqrt{2mE}) + B \delta(p + \sqrt{2mE}) \quad \sqrt{2mE} = k\hbar \quad \checkmark$$

2) $V(x)=0$ & $V'(x)=V_0$ describes the same physics (free particle)

In momentum space

$$V'(p) = \frac{1}{\sqrt{2\pi\hbar}} \int dx V'(x) e^{-ipx/\hbar} = \frac{V_0}{\sqrt{2\pi\hbar}} \int dx e^{-ipx/\hbar} = \sqrt{2\pi\hbar} V_0 \delta(p)$$

$$\Rightarrow V(p)=0 \text{ \& \; } V'(p) = V_0 \delta(p) \text{ describe the same physics}$$

Further, the Schrödinger eq. in momentum space reads

$$+\frac{p^2}{2m} \psi(p) + \int dp' \frac{1}{\sqrt{2\pi\hbar}} V(p-p') \psi(p') = E \psi(p)$$

$$\Leftrightarrow +\frac{p^2}{2m} \psi(p) + V_0 \psi(p) = E \psi(p)$$

which can be compared directly with the one in position-sp.

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V_0 \psi(x) = E \psi(x)$$

3) The momentum operator: $\hat{p}$

With the available tools, we can determine the form of the mom. op. in the coord. rep.

$$\langle x | \hat{p} | \psi \rangle = \int dx' \langle x | \hat{p} | x' \rangle \langle x' | \psi \rangle$$

$$= \int dx' \int dp \int dp' \underbrace{\langle x | p \rangle}_{1} \underbrace{\langle p | \hat{p} | p' \rangle}_{p} \underbrace{\langle p' | x' \rangle}_{\psi(x')}$$

$$= \int dx' \int dp \int dp' \langle x | p \rangle p \delta(p - p') \langle p' | x' \rangle \psi(x')$$

$$= \int dx' \int dp \langle x | p \rangle p \langle p | x' \rangle \psi(x')$$

$$p \langle p | x' \rangle = \frac{p}{\sqrt{2\pi\hbar}} e^{-ipx'/\hbar} = i\hbar \frac{1}{\sqrt{2\pi\hbar}} \frac{d}{dx'} e^{-ipx'/\hbar} = i\hbar \frac{d}{dx'} \langle p | x' \rangle$$

$$\langle x | \hat{p} | \psi \rangle = \int dx' i\hbar \frac{d}{dx'} \left(\int dp \langle x | p \rangle \langle p | x' \rangle \right) \psi(x')$$

$$= \int dx' i\hbar \frac{d}{dx'} \left(\delta(x - x') \right) \psi(x')$$

$$\stackrel{\text{by parts}}{=} - \int dx' i\hbar \delta(x - x') \frac{d}{dx'} \psi(x') = -i\hbar \frac{d}{dx} \psi(x)$$

It looks nicer if we use

$$p \langle x | p \rangle = \frac{p}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} = -i\hbar \frac{1}{\sqrt{2\pi\hbar}} \frac{d}{dx} e^{ipx/\hbar} = -i\hbar \frac{d}{dx} \langle x | p \rangle$$

$$\Rightarrow \langle x | \hat{p} | \psi \rangle = -i\hbar \frac{d}{dx} \int dx' \left(\int dp \langle x | p \rangle \langle p | x' \rangle \right) \psi(x')$$

$$= -i\hbar \frac{d}{dx} \psi(x)$$

§ 6. Quantum systems

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~ 6.1. The harmonic oscillator

The potential of the harmonic oscillator ~~in $D=1$~~ is given by

$$\hat{V}(\hat{x}) = \frac{1}{2} m \omega^2 \hat{x}^2$$



like a well $\rightarrow$ quantization

The ^{static} Schrödinger eq. in position rep. is

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \frac{1}{2} m \omega^2 x^2 \psi = E \psi$$

Let's choose adimensional variables

$$\tilde{x} \equiv \sqrt{\frac{m\omega}{\hbar}} x$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{m\omega}{\hbar} \frac{\partial^2 \psi}{\partial \tilde{x}^2} + \frac{1}{2} \hbar \omega \frac{\hbar}{m\omega} \tilde{x}^2 \psi = E \psi$$

$$\Leftrightarrow \boxed{\psi'' - \tilde{x}^2 \psi = \frac{-2E}{\hbar\omega} \psi}$$

Propose the ansatz (motivated because $\psi'' - \tilde{x}^2 \psi = 0 \Rightarrow \psi = e^{-\frac{1}{2}\tilde{x}^2}$)

$$\psi(\tilde{x}) = A e^{-\frac{1}{2}\tilde{x}^2} u(\tilde{x})$$

$$\Rightarrow A e^{-\frac{1}{2}\tilde{x}^2} \left(u'' - 2\tilde{x}u' - u + \tilde{x}^2 u - \tilde{x}u + \frac{2E}{\hbar\omega} u \right) = 0$$

$$\Leftrightarrow u'' - 2\tilde{x}u' + (\lambda + 1)u = 0 \quad \lambda \equiv \frac{2E}{\hbar\omega} \quad (*)$$

There is a very similar differential equation (see app. A2 of De la Peña.)
(also §13 of Arfken. Math. Meth. for Physicist)

$$u'' - 2\tilde{x}u' + 2nu = 0 \quad n \in \mathbb{Z}$$

whose solutions are the Hermite polynomials

$$u(\tilde{x}) = H_n(\tilde{x}) = (-1)^n e^{\tilde{x}^2} \frac{d^n}{d\tilde{x}^n} e^{-\tilde{x}^2} \quad n \in \mathbb{Z}$$

Thus in fact, (*) only has solution iff

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$$\lambda - 1 = 2n \Leftrightarrow \frac{2E}{\hbar\omega} = 2n + 1$$

$$\Rightarrow \boxed{E_n = \frac{\hbar\omega}{2} (2n+1) = \hbar\omega(n + \frac{1}{2})} \rightsquigarrow \underline{\underline{E_0 \neq 0}}$$

To each energy-level, there's a solution of the form

$$\boxed{\psi_n(\tilde{x}) = A_n e^{-\frac{1}{2}\tilde{x}^2} H_n(\tilde{x})} \quad \leftarrow \text{Hermite polynomials}$$

where

$$H_0 = 1$$

$$H_1 = 2\tilde{x}$$

$$H_2 = 4\tilde{x}^2 - 2$$

$$H_3 = 8\tilde{x}^3 - 12\tilde{x}$$

$$\tilde{x} = \sqrt{\frac{m\omega}{\hbar}} x$$

The ~~normalization~~ constant A_n is obtained by using

$$\begin{aligned} \int_{-\infty}^{\infty} \psi_n^*(x) \psi_n(x) dx &= \sqrt{\frac{\hbar}{m\omega}} \int_{-\infty}^{\infty} \psi_n^*(\tilde{x}) \psi_n(\tilde{x}) d\tilde{x} \\ &= \sqrt{\frac{\hbar}{m\omega}} |A_n|^2 \int_{-\infty}^{\infty} d\tilde{x} e^{-\tilde{x}^2} H_n^2(\tilde{x}) \\ &= \sqrt{\frac{\hbar}{m\omega}} |A_n|^2 \int_{-\infty}^{\infty} d\tilde{x} e^{-\tilde{x}^2} H_n(\tilde{x}) e^{\tilde{x}^2} (-1)^n \frac{d^n}{d\tilde{x}^n} e^{-\tilde{x}^2} = 1 \end{aligned}$$

This integral can be shown to be (see §11 De La Peña)

$$\sqrt{\frac{\hbar}{m\omega}} |A_n|^2 \int_{-\infty}^{\infty} d\tilde{x} e^{-\tilde{x}^2} \underbrace{\frac{d^n H_n}{d\tilde{x}^n}}_{2^n n!} = \sqrt{\frac{\hbar}{m\omega}} |A_n|^2 2^n n! \underbrace{\int_{-\infty}^{\infty} d\tilde{x} e^{-\tilde{x}^2}}_{\sqrt{\pi}} \stackrel{!}{=} 1$$

$$\Leftrightarrow \boxed{A_n = \left(\frac{\sqrt{\hbar\pi}}{\sqrt{m\omega}} 2^n n! \right)^{-1/2}}$$

(Can use orthogonality $\int_{-\infty}^{\infty} dx e^{-x^2} H_n H_m = 2^n n! \sqrt{\pi} \delta_{n,m}$)

§ ~~Harmonic Oscillator~~ Harmonic Oscillator in operator method

(Sakurai §2.3)

$$V(x) = \frac{1}{2} m \omega^2 x^2$$

$$E|\psi\rangle = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \frac{1}{2} m \omega^2 x^2 \psi = \frac{\hat{p}^2}{2m} |\psi\rangle + \frac{1}{2} m \omega^2 \hat{x}^2 |\psi\rangle$$

Useful to define

$$\hat{a} \equiv \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i\hat{p}}{m\omega} \right)$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger)$$

$$\hat{a}^\dagger \equiv \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i\hat{p}}{m\omega} \right)$$

$$\hat{p} = i\sqrt{\frac{m\hbar\omega}{2}} (\hat{a}^\dagger - \hat{a})$$

Notice that

$$\begin{aligned} [\hat{a}, \hat{a}^\dagger] &= \frac{m\omega}{2\hbar} \left([\hat{x}, \frac{-i\hat{p}}{m\omega}] + [\frac{i\hat{p}}{m\omega}, \hat{x}] \right) \\ &= \frac{1}{2\hbar} \left(\underbrace{-i[\hat{x}, \hat{p}]}_{i\hbar} + \underbrace{i[\hat{p}, \hat{x}]}_{-i\hbar} \right) = 1 \end{aligned}$$

$\hbar + \hbar$

~~Define~~ Define a so-called "number op."

$$\hat{N} \equiv \hat{a}^\dagger \hat{a} = \frac{m\omega}{2\hbar} \left(\hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} \right) + \frac{i}{2\hbar} [\hat{x}, \hat{p}]$$

$$= \frac{1}{\hbar\omega} \underbrace{\left(\frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2 \right)}_{\hat{H}} + \frac{1}{2}$$

$$\Rightarrow \hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right)$$

The eigenstates of $\hat{N}$ are defined as

$$\hat{N}|n\rangle = n|n\rangle \quad \text{where } n \text{ is an arbitrary number} \\ \text{(later } n > 0, n \in \mathbb{Z} \text{) to be shown}$$

$\Rightarrow$ the $\{|n\rangle\}$ are also eigenstates of $\hat{H}$

$$\hat{H}|n\rangle = \hbar\omega(n + \frac{1}{2})|n\rangle$$

$$\Rightarrow \boxed{E_n = \hbar\omega(n + \frac{1}{2})}$$

Notice now that

$$[\hat{N}, \hat{a}^\dagger] = [\hat{a}^\dagger \hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger [\hat{a}, \hat{a}^\dagger] + \hat{a}^\dagger \hat{a} \hat{a}^\dagger - \hat{a}^\dagger \hat{a}^\dagger \hat{a} = \hat{a}^\dagger$$

$$[\hat{N}, \hat{a}] = [\hat{a}^\dagger \hat{a}, \hat{a}] = \hat{a}^\dagger [\hat{a}, \hat{a}] + [\hat{a}^\dagger, \hat{a}] \hat{a} = -\hat{a}$$

$$\Rightarrow \hat{N} \hat{a}^\dagger |n\rangle = ([\hat{N}, \hat{a}^\dagger] + \hat{a}^\dagger \hat{N}) |n\rangle \\ = (\hat{a}^\dagger + \hat{a}^\dagger n) |n\rangle = (n+1) \hat{a}^\dagger |n\rangle$$

$$\hat{N} \hat{a} |n\rangle = ([\hat{N}, \hat{a}] + \hat{a} \hat{N}) |n\rangle = (n-1) \hat{a} |n\rangle$$

$\Rightarrow$ Eigenvalue of state $\hat{a}^\dagger |n\rangle$ is $n+1$
 $\hat{a} |n\rangle$ is $n-1$

BUT we know that

$$\hat{N}(c|n+1\rangle) = (n+1)(c|n+1\rangle) \quad c, d \in \mathbb{R}$$

$$\hat{N}(d|n-1\rangle) = (n-1)(d|n-1\rangle)$$

$$\Rightarrow \hat{a}^\dagger |n\rangle = c|n+1\rangle \quad \rightsquigarrow \hat{a}^\dagger \text{ is a creation op.}$$

$$\hat{a} |n\rangle = d|n-1\rangle \quad \rightsquigarrow \hat{a} \text{ is an annihilation op.}$$

We can compute c & d :

$$\langle n | \hat{a}^\dagger \hat{a} | n \rangle = |d|^2 \underbrace{\langle n+1 | n+1 \rangle}_{=1 \text{ assume}}$$

$$\langle n | \hat{N} | n \rangle$$

$$\langle n | n \rangle = 1$$

$$\Rightarrow \boxed{d = \sqrt{n}} \quad \& \quad \boxed{n \geq 0}$$

$$a^\dagger a \quad (55) \quad (56)$$

$$\hat{x}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$|d|^2 \langle n+1 | n+1 \rangle = \langle n | \hat{a}^\dagger \hat{a}^\dagger | n \rangle = \langle n | [\hat{a}, \hat{a}^\dagger] + \hat{a}^\dagger \hat{a} | n \rangle = \langle n | 1 + \hat{N} | n \rangle$$

$$= (n+1) \underbrace{\langle n | n \rangle}_{=1}$$

$$\Rightarrow \boxed{d = \sqrt{n+1}}$$

Since $\boxed{n=0}$ is the smallest possible value & all states created by $\hat{a}^\dagger$ differ by a unit, $n \in \mathbb{Z}$. Then the ground state (state of lowest possible energy) is $|n=0\rangle$

$$\hat{H}|0\rangle = \frac{1}{2}\hbar\omega|0\rangle$$

has non-vanishing energy!! Notice that $\hat{a}|0\rangle=0$ is a reasonable way to define the ground state!

We can obtain an arbitrary state $|n\rangle$ by applying creation ops. on $|0\rangle$:

$$|1\rangle = \hat{a}^\dagger |0\rangle$$

$$|2\rangle = \frac{\hat{a}^\dagger}{\sqrt{2}} |1\rangle = \frac{(\hat{a}^\dagger)^2}{\sqrt{2}} |0\rangle$$

$$|3\rangle = \frac{\hat{a}^\dagger}{\sqrt{3}} |2\rangle = \frac{(\hat{a}^\dagger)^3}{\sqrt{3!}} |0\rangle$$

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle$$

The states $\{|n\rangle\}$ build an orthonormal basis: $\langle n | m \rangle = \delta_{n,m}$ (see next p.)

$$\Rightarrow \left. \begin{aligned} \langle m | \hat{a} | n \rangle &= \sqrt{n} \delta_{m,n-1} \\ \langle m | \hat{a}^\dagger | n \rangle &= \sqrt{n+1} \delta_{m,n+1} \end{aligned} \right\} \text{non-diagonal ops.}$$

$$\Rightarrow \langle m | \hat{x} | n \rangle = \sqrt{\frac{\hbar}{2m\omega}} \langle m | \hat{a} + \hat{a}^\dagger | n \rangle = \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n} \delta_{m,n-1} + \sqrt{n+1} \delta_{m,n+1})$$

$$\langle m | \hat{p} | n \rangle = i\sqrt{\frac{m\hbar\omega}{2}} (\sqrt{n+1} \delta_{m,n+1} - \sqrt{n} \delta_{m,n-1})$$

$$m \neq n$$

$$a) \quad m > n$$

~~for~~

$$|m\rangle = \frac{(\hat{a}^\dagger)^m}{\sqrt{m!}} |0\rangle = \frac{(\hat{a}^\dagger)^{n+m'}}{\sqrt{(n+m')!} \sqrt{n!}} |0\rangle$$

$$= \frac{(\hat{a}^\dagger)^{m'}}{\sqrt{(n+m')!}} \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle$$

$$= \sqrt{\frac{n!}{(n+m')!}} (\hat{a}^\dagger)^{m'} |n\rangle$$

$$\langle m | n \rangle = \sqrt{\frac{n!}{(n+m')!}} \underbrace{\langle n | (\hat{a}^\dagger)^{m'} | n \rangle}_{=0} = 0$$

b) $n \geq m$ is analogous

$$\Rightarrow \langle n | m \rangle = \delta_{n,m}$$

$\rightarrow \hat{x}$ & $\hat{p}$ are also non-diagonal operators in the basis $\{|n\rangle\}$ of eigenstates of $\hat{H}$. (Recall: $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger)$, $\hat{p} = i\sqrt{\frac{m\hbar\omega}{2}} (\hat{a}^\dagger - \hat{a})$)

We can now build the states $\{|n\rangle\}$ in the position space:

$$\hat{a}|0\rangle = 0$$

$$\Rightarrow \langle x | \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i\hat{p}}{m\omega} \right) | 0 \rangle = 0$$

$$\Leftrightarrow \int dx' \langle x | \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i\hat{p}}{m\omega} \right) | x' \rangle \langle x' | 0 \rangle = 0$$

$$\Leftrightarrow \int dx' \delta(x-x') \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{i}{m\omega} \left(-i\hbar \frac{d}{dx} \right) \right) \langle x' | 0 \rangle = 0$$

$$\Leftrightarrow \left(x + \frac{\hbar}{m\omega} \frac{d}{dx} \right) \langle x | 0 \rangle = 0$$

$$\Leftrightarrow \frac{d\langle x | 0 \rangle}{dx} = -\frac{m\omega}{\hbar} x \langle x | 0 \rangle$$

$$\Leftrightarrow \frac{d\langle x|0\rangle}{\langle x|0\rangle} = -\frac{m\omega}{\hbar} x dx \Rightarrow \ln \langle x|0\rangle = -\frac{1}{2} \frac{m\omega}{\hbar} x^2 = -\frac{1}{2} \tilde{x}^2, \quad \tilde{x} = \sqrt{\frac{m\omega}{\hbar}} x$$

$$\Leftrightarrow \langle x|0\rangle = A_0 e^{-\frac{1}{2} \tilde{x}^2}$$

Normalizing as before

$$\langle x|0\rangle = \left(\frac{\hbar \pi}{m\omega}\right)^{-1/4} e^{-\frac{1}{2} \tilde{x}^2} = \psi_0(\tilde{x}) \quad \checkmark$$

The excited states $\{|n\rangle, n>0\}$ can be computed too

$$\langle x|1\rangle = \langle x|\hat{a}^\dagger|0\rangle = \sqrt{\frac{m\omega}{2\hbar}} \langle x|\hat{x} - \frac{i\hat{p}}{m\omega}|0\rangle$$

$$= \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{\hbar}{m\omega} \frac{d}{dx}\right) \langle x|0\rangle$$

$$\langle x|2\rangle = \langle x|\frac{(\hat{a}^\dagger)^2}{\sqrt{2}}|0\rangle = \frac{1}{\sqrt{2}!} \left(\sqrt{\frac{m\omega}{2\hbar}}\right)^2 \left(x - \frac{\hbar}{m\omega} \frac{d}{dx}\right)^2 \langle x|0\rangle$$

$$\langle x|3\rangle = \langle x|\frac{(\hat{a}^\dagger)^3}{\sqrt{3}!}|0\rangle = \frac{1}{\sqrt{3}!} \left(\sqrt{\frac{m\omega}{2\hbar}}\right)^3 \left(x - \frac{\hbar}{m\omega} \frac{d}{dx}\right)^3 \langle x|0\rangle$$

⋮

$$\begin{aligned} \langle x|n\rangle &= \frac{1}{\sqrt{n! 2^n}} \frac{1}{\left(\sqrt{\frac{\hbar}{m\omega}}\right)^n} \left(\frac{\hbar \pi}{m\omega}\right)^{-1/4} \underbrace{\left(x - \frac{\hbar}{m\omega} \frac{d}{dx}\right)^n}_{\left(\sqrt{\frac{\hbar}{m\omega}} \tilde{x} - \frac{\hbar}{m\omega} \sqrt{\frac{m\omega}{\hbar}} \frac{d}{d\tilde{x}}\right)^n} e^{-\frac{1}{2} \tilde{x}^2} \\ &= \frac{1}{\sqrt{n! 2^n}} \left(\frac{\hbar \pi}{m\omega}\right)^{-1/4} \left(\tilde{x} - \frac{d}{d\tilde{x}}\right)^n e^{-\frac{1}{2} \tilde{x}^2} \end{aligned}$$

$$= \frac{1}{\sqrt{n! 2^n}} \left(\frac{\hbar \pi}{m\omega}\right)^{-1/4} \left(\tilde{x} - \frac{d}{d\tilde{x}}\right)^n e^{-\frac{1}{2} \tilde{x}^2}$$

$$= \frac{1}{\sqrt{n! 2^n}} \left(\frac{\hbar \pi}{m\omega}\right)^{-1/4} H_n(\tilde{x}) e^{-\frac{1}{2} \tilde{x}^2} = \psi_n(x) \text{ as before!}$$

$$H_n(\tilde{x}) = e^{\frac{1}{2} \tilde{x}^2} \left(\tilde{x} - \frac{d}{d\tilde{x}}\right)^n e^{-\frac{1}{2} \tilde{x}^2} \text{ another expression}$$